CognEn v1.0 — Cognitive Engagement Analyzer (Browser)

> ***Single-student, webcam-based real-time attention monitoring*** *For the multi-person CCTV server version, see* ***v2.0 README****.*

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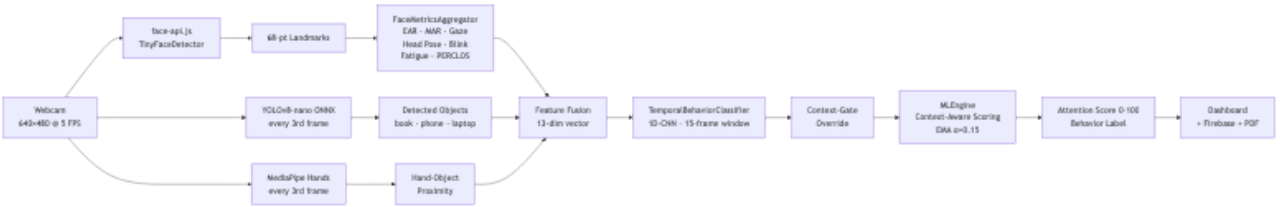
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1. Overview

CognEn v1.0 is a **browser-based** cognitive engagement analyzer built on Next.js 15. It captures a webcam feed at 640×480 @ 5 FPS, runs face detection via **face-api.js TinyFaceDetector**, extracts 68 facial landmarks, and computes a real-time **attention score (0–100)** using a weighted multi-feature ML engine with EMA temporal smoothing.

The upgraded v1.0 adds **context-aware scoring**: YOLOv8-nano object detection, MediaPipe hand tracking, and a 15-frame sliding-window temporal classifier distinguish reading a book (attentive) from using a phone (distracted) — eliminating the uniform downward-gaze penalty of the original system.



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2. Key Features

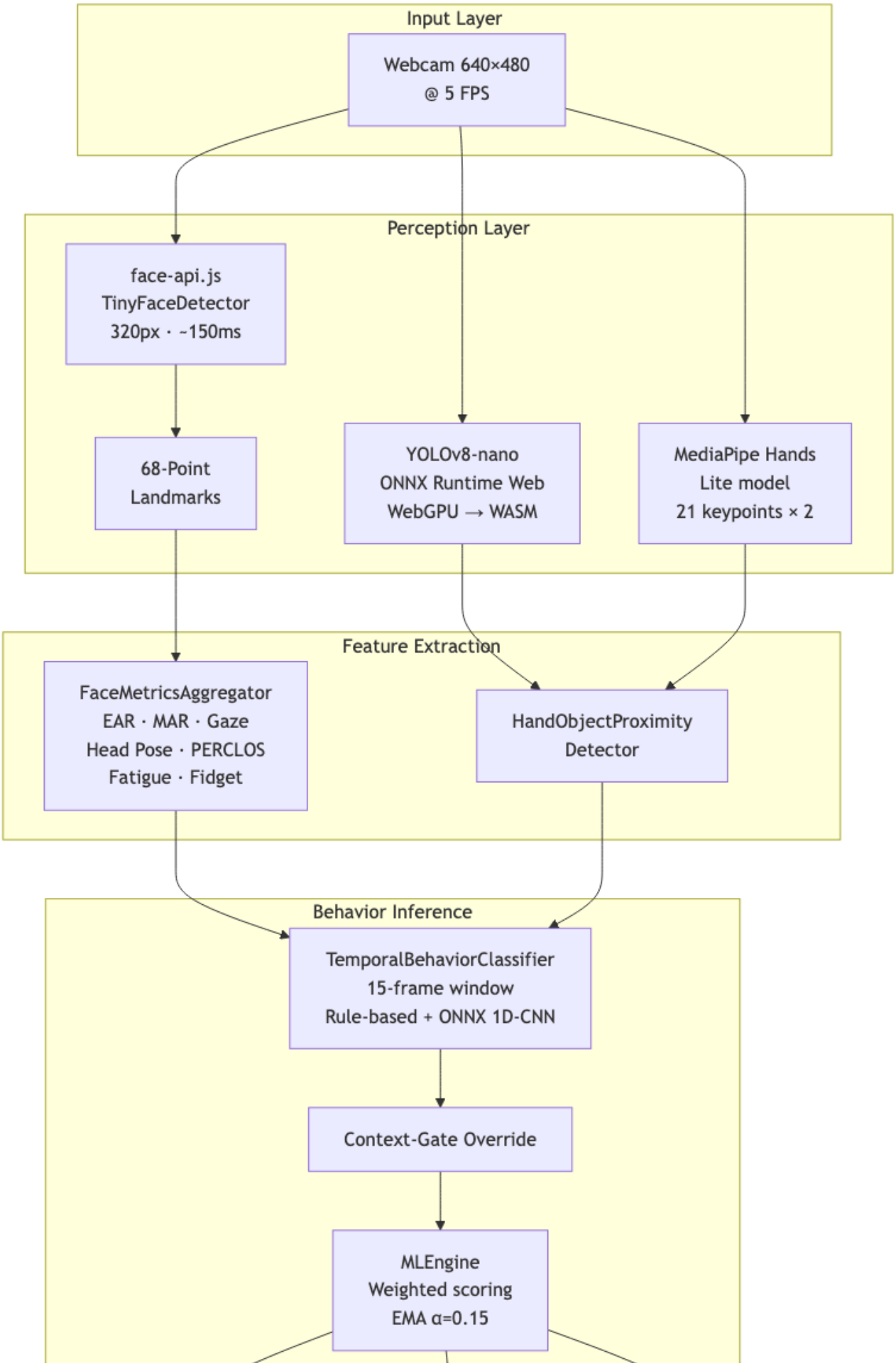
Feature	Description
Context-Aware Attention	Distinguishes phone use from book reading via YOLOv8-nano + MediaPipe hand tracking
Temporal Classification	1D-CNN sliding-window (15 frames / 3 s) with 6 behavior classes

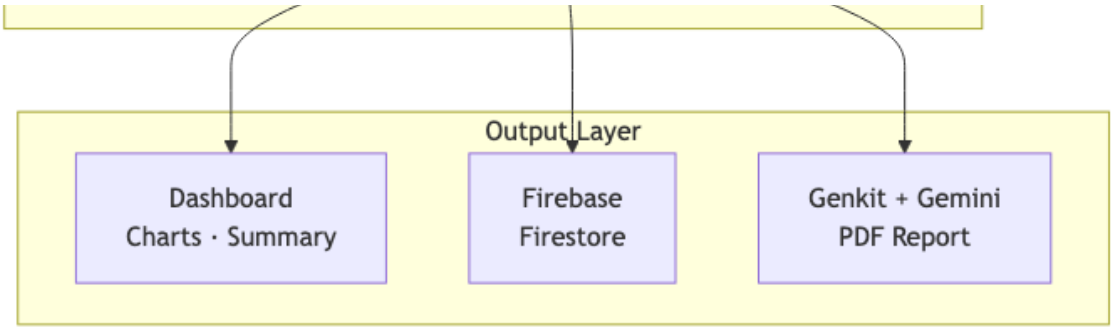
Context-Gate Override	Hard rule post-processor resolving phone/book ambiguity below confidence threshold
12+ Biometric Features	EAR, MAR, PERCLOS, gaze deviation, head pose, blink rate, fatigue index, fidget score
Live Dashboard	Real-time line chart, pie chart, multitasking counter, session summary
PDF Reports	AI-generated session reports via Google Genkit + Gemini
Chrome Extension	Manifest V3, works inside Google Meet / Zoom / Teams via offscreen document
Firebase Auth	Email/password + Google Sign-in, Firestore session storage

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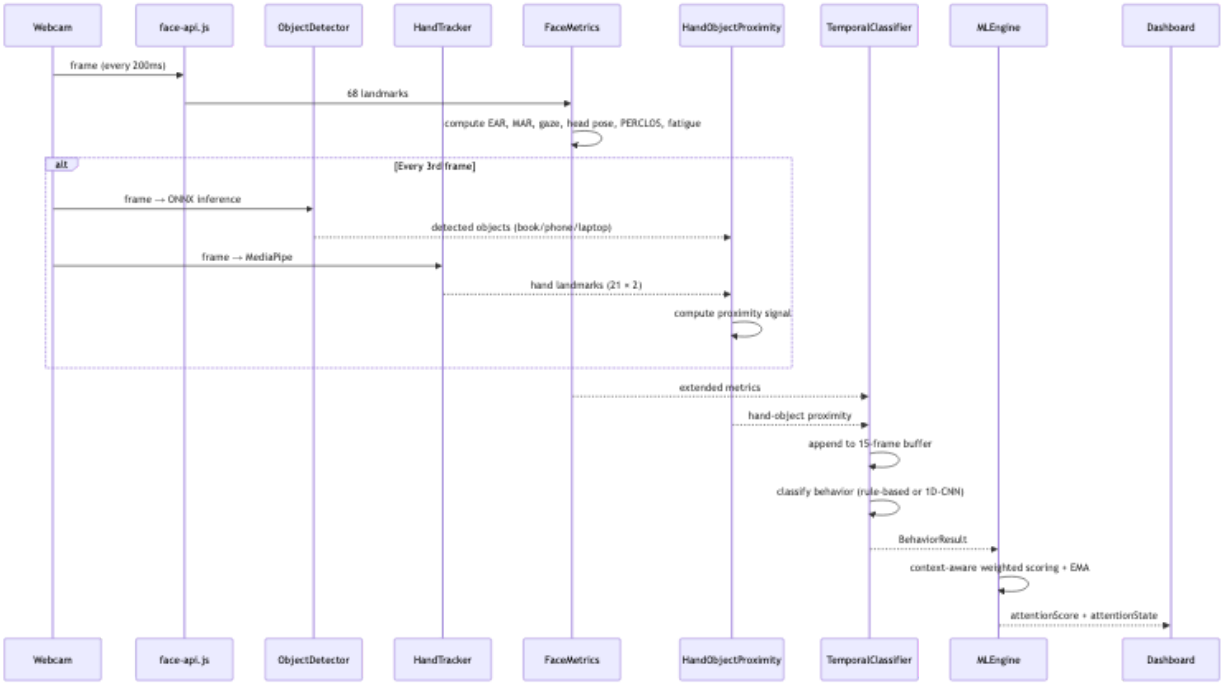
3. System Architecture

3.1 High-Level Data Flow


Input Layer
Webcam 640×480
@ 5 FPS
Perception Layer
face-api.js
TinyFaceDetector
320px · ~150ms
68-Point
Landmarks
YOLOv8-nano
ONNX Runtime Web
WebGPU → WASM
MediaPipe Hands
Lite model
21 keypoints × 2
Feature Extraction
FaceMetricsAggregator
EAR · MAR · Gaze
Head Pose · PERCLOS
Fatigue · Fidget
HandObjectProximity
Detector
Behavior Inference
TemporalBehaviorClassifier
15-frame window
Rule-based + ONNX 1D-CNN
Context-Gate Override
MLEngine
Weighted scoring
EMA α=0.15



3.2 Sequence Diagram



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4. ML Pipeline & Algorithms

4.1 Two Scoring Modes

The MLEngine automatically selects its scoring mode:

Mode	Condition	Features	Weights
LEGACY	`detectedObjects === undefined`	13 features incl. uniform `pitchDeviation`	`FEATURE_WEIGHTS_LEGACY`
UPGRADED	`detectedObjects !== undefined`	16 features: replaces `pitchDeviation` with `contextAwarePitch`, `objectContext`, `handObjectProximity`, `temporalPattern`	`FEATURE_WEIGHTS_V1_UPGRADED`

4.2 Upgraded Feature Weights

Feature	Weight	Role
gazeDeviation	0.12	Primary attention signal

contextAwarePitch	0.12	Pitch gated by object context
objectContext	0.10	What objects are visible
headPoseStability	0.10	Overall engagement
ear	0.10	Eye openness
perclos	0.08	Drowsiness
yawDeviation	0.08	Looking left/right
fatigueIndex	0.07	Accumulated fatigue
blinkRate	0.05	Eye strain indicator
handObjectProximity	0.05	Hand-object interaction
fidgetScore	0.04	Restlessness
temporalPattern	0.03	Temporal consistency bonus
mouthOpenness	0.02	Yawning detection
faceDistance	0.02	Proximity to camera
facePresence	0.02	Face in frame

4.3 Score Computation

```
For each feature:
  score_i = scoreFeature(name, value)    // → 0–100
  quality_i = getQuality(name, value)    // → 0–1

Weighted combination:
  raw = Σ(score_i × weight_i × quality_i) / Σ(weight_i × quality_i)

EMA smoothing:
  smoothed = α × raw + (1 − α) × prev_smoothed    (α = 0.15)

Classification:
  ≥ 70 → Attentive
  ≥ 40 → Distracted
  < 40 → Multitasking
```

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5. Feature Extraction — Mathematical Foundations

5.1 Eye Aspect Ratio (EAR)

```
EAR = (‖p2−p6‖ + ‖p3−p5‖) / (2 × ‖p1−p4‖)
```

EAR Value	Interpretation
0.30–0.40	Fully open
0.25–0.30	Normal
0.21–0.25	Partially closed

< 0.21	Blink / closed

Blink detection: EAR drops below 0.21 then rises back, duration 50–500 ms.

5.2 PERCLOS (Percentage Eye Closure)

```
PERCLOS = (frames with EAR < 0.21) / (total frames in last 60s) × 100%
```

PERCLOS	Interpretation
< 15%	Alert
15–30%	Mildly fatigued
> 30%	Drowsy

5.3 Mouth Aspect Ratio (MAR)

```
MAR = ‖top_lip − bottom_lip‖ / ‖left_corner − right_corner‖
```

MAR	Interpretation
< 0.10	Mouth closed
0.10–0.25	Speaking
> 0.25	Yawning

5.4 Head Pose Estimation

```
Yaw   = ((nose.x − eye_midpoint.x) / eye_distance) × 50°
Pitch = ((nose.y − eye_midpoint.y) / nose_chin_distance) × 50°
Roll  = atan2(right_eye.y − left_eye.y, right_eye.x − left_eye.x) × (180/π)
```

Stability Score (variance over 30-second window):

```
Stability = 100 − (σ²_yaw + σ²_pitch + 0.5 × σ²_roll) / 2.5 / 10
```

5.5 Gaze Direction

```
Gaze_X = ((iris_center.x − eye_center.x) / eye_width − 0.5) × 2
Gaze_Y = (eye_center.y − iris_center.y) / eye_height
Deviation = √(Gaze_X² + Gaze_Y²)
```

Deviation	Interpretation
< 0.15	Looking at screen
0.15–0.30	Slight glance
> 0.30	Looking away

5.6 Fatigue Index

Composite of three sub-scores (0–100):

```
PERCLOS_score   = clamp(avg_PERCLOS × 2, 0, 40)
Blink_score     = penalty if outside 15–20 bpm range (max 30)
EAR_trend_score = penalty if avg_EAR < 0.25 (max 30)
Fatigue_Index   = clamp(PERCLOS_score + Blink_score + EAR_trend_score, 0, 100)
```

5.7 Fidget Score

```
Movement = Σ|Δyaw| + |Δpitch| + |Δroll| over recent frames
Fidget   = clamp(avg_movement × 5, 0, 100)
```

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6. Object Detection Module (YOLOv8-nano)

Model: YOLOv8-nano (3.2M params) via ONNX Runtime Web (WebGPU → WASM fallback)

Property	Value
Input	`[1, 3, 640, 640]` float32, NCHW, normalized [0,1]
Output	`[1, 84, 8400]` (cx cy w h + 80 COCO class scores)
Confidence threshold	0.25
NMS IoU threshold	0.45
Inference cadence	Every 3rd frame
Target classes	laptop(63), mouse(64), keyboard(66), phone(67), book(73)

Post-processing: Class-aware NMS — only suppresses overlapping boxes within the same class.

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7. Hand Tracking Module (MediaPipe Hands)

Property	Value
Model	MediaPipe Hands Lite (complexity 0)
Max hands	2
Landmarks	21 per hand (normalized [0,1])
Cadence	Every 3rd frame (same as ObjectDetector)
Min detection confidence	0.5

HandObjectProximityDetector fuses hand centroids with object bounding boxes:

- Proximity threshold: 0.20 (normalized Euclidean distance)
- Priority order: `phone > book/notebook > pen`
- Output labels: `holding_phone`, `holding_book`, `holding_pen`, `hands_on_desk`, `hands_not_visible`

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8. Temporal Behavior Classifier (1D-CNN)

8.1 Feature Vector (13 dimensions per frame)

Index	Feature	Range

0	gaze_x	[-1, 1]
1	gaze_y	[-1, 1]
2	pitch_norm (pitch/90)	[-1, 1]
3	yaw_norm (yaw/90)	[-1, 1]
4	ear	[0, 1]
5	perclos/100	[0, 1]
6	head_pose_stability/100	[0, 1]
7	has_book (one-hot)	{0, 1}
8	has_phone (one-hot)	{0, 1}
9	has_laptop (one-hot)	{0, 1}
10	has_none (one-hot)	{0, 1}
11	fidget_score/100	[0, 1]
12	proximity_code	[0, 1]

8.2 Behavior Classes

Class	Attention Score	Description
`attentive_screen`	85	Watching lecture / screen
`attentive_reading`	80	Reading book / notes
`distracted`	45	Looking around, talking
`using_phone`	25	Phone distraction
`drowsy`	20	Head down + low EAR
`absent`	0	Face not in frame

8.3 Two-Tier Classification

1. **Rule-based fallback** — always available, used when < 15 frames buffered or no ONNX model
2. **ONNX 1D-CNN** — trained model loaded via `loadModel()`, input `[1, 15, 13]`

8.4 Rule-Based Decision Tree

```
1. face_area < 0.005         → absent     (conf 1.0)
2. EAR < 0.22 AND PERCLOS > 20 → drowsy   (conf 0.85)
3. phone + (holding OR looking down) → using_phone (conf 0.80)
4. book + (holding OR looking down) → attentive_reading (conf 0.75)
5. gaze_deviation < 0.3 AND |pitch| < 15° → attentive_screen (conf 0.70)
6. else → distracted (conf 0.60)
```

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9. Context-Aware Scoring Engine

9.1 Context-Aware Pitch Score

Replaces the uniform pitch penalty:

```
if pitch < −15°:
    if phone detected or holding_phone → 20 (distracted)
    if book detected or holding_book  → 85 (attentive)
    if EAR < 0.22 and PERCLOS > 20   → 15 (drowsy)
    else                              → 40 (unknown)
else:
    standard pitch scoring based on deviation from optimal ±15°
```

9.2 Object Context Score

```
No objects       → 65 (neutral)
Phone visible    → 25
Book + pitch<−10 → 80
Book (other)     → 70
Laptop + |pitch|<15 → 80
Other            → 65
```

9.3 Hand Proximity Encoding

Proximity	Score
holding_book	80
holding_pen	75
hands_on_desk	65
hands_not_visible	60
holding_phone	20

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10. Session Analytics Engine

Uses **linear regression** for trend detection, sliding windows for period identification.

Algorithm	Purpose
Linear regression (OLS)	Attention trend (slope + R^2)
Period detection	Find contiguous high (≥75) / low (≤45) periods
Fatigue onset detection	Sustained declining slope (< -0.5) + R^2 > 0.4 + avg < 60
Anomaly detection	Z-score > 2.5 σ from mean; sudden drops > 30 points
Optimal window	Sliding window finding highest average attention period

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11. Chrome Extension

Manifest V3 extension with dual-mode operation:

```
flowchart LR
    POP["popup.js<br/>(UI)"] --> BG["background.js<br/>(Service Worker)"]
    BG --> CT["content.js<br/>(Meet Inject)"]
```

```
    CT --> OS["offscreen.js<br/>(Standalone)"]
    OS --> MLC["ml-core.js<br/>(ML Engine)"]
```

Mode	When	How
Meet Mode	On Google Meet tab	Injects into page, reads existing video element
Standalone	Not on Meet	Uses offscreen document with own camera access

Score Interpretation: 70–100 (Green/Attentive) · 40–69 (Orange/Distracted) · 0–39 (Red/Multitasking)

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12. Tech Stack

Layer	Technology
Framework	Next.js 15 (App Router, Turbopack)
Face detection	face-api.js TinyFaceDetector (320px, ~150ms)
Landmarks	face-api.js 68-point model (iBUG convention)
Object detection	YOLOv8-nano ONNX via `onnxruntime-web` (WebGPU → WASM)
Hand tracking	MediaPipe Hands Lite (`@mediapipe/hands`)
Temporal classifier	Rule-based + ONNX 1D-CNN
Auth / Storage	Firebase Auth + Firestore
AI report	Google Genkit + Gemini (PDF generation)
Charts	Recharts
UI components	Radix UI + Tailwind CSS
Extension	Chrome MV3 + offscreen document

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13. Repository Structure

```
v1.0/
├── src/
│   ├── lib/
│   │   ├── types.ts              # Core types: ExtendedFaceMetrics, BehaviorResult, etc.
│   │   ├── face-geometry.ts      # EAR/MAR, BlinkDetector, HeadPoseTracker,
│   │   │                         #   FatigueCalculator, HandObjectProximityDetector
│   │   ├── ml-engine.ts          # MLEngine: context-aware scoring, FEATURE_WEIGHTS
│   │   ├── object-detector.ts    # YOLOv8-nano ONNX wrapper (browser)
│   │   ├── hand-tracker.ts       # MediaPipe Hands wrapper (browser)
│   │   ├── temporal-classifier.ts # 1D-CNN sliding-window classifier (browser)
│   │   ├── analytics.ts          # SessionAnalyticsEngine
│   │   └── firebase.ts           # Firebase config
│   ├── components/
│   │   ├── dashboard/            # Webcam panel, charts, session summary
│   │   ├── auth-form.tsx         # Login / signup
│   │   └── ui/                   # Radix UI primitives
│   ├── app/
│   │   ├── page.tsx              # Landing / auth page
```

```
│   │   ├── dashboard/            # Main dashboard route
│   │   └── globals.css           # Global styles
│   ├── ai/flows/                 # Genkit flows (session summary, PDF report)
│   └── hooks/                    # React hooks
├── extension/
│   ├── lib/ml-core.js            # JS port: MLEngineV2, TemporalBehaviorClassifier
│   ├── offscreen.js              # Offscreen document: webcam + face-api.js
│   ├── background.js             # Service worker
│   ├── content.js                # Injects attention widget into Meet/Zoom/Teams
│   └── manifest.json             # Chrome MV3 manifest
├── public/
│   ├── models/                   # face-api.js weights
│   └── onnx/                     # onnxruntime-web WASM binaries
├── scripts/
│   └── setup-models.sh           # Copies WASM from node_modules
├── package.json
├── next.config.ts
├── tailwind.config.ts
└── tsconfig.json
```

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14. Setup & Installation

Prerequisites

- Node.js ≥ 20
- Firebase project (Auth + Firestore)
- Google AI API key (Genkit / Gemini)

Steps

```
# 1. Install dependencies
cd v1.0 && npm install

# 2. Environment variables
cp .env.example .env.local
# Edit with Firebase + Genkit keys (see §15)

# 3. ONNX Runtime WASM binaries
bash scripts/setup-models.sh

# 4. YOLOv8-nano model (on a machine with Ultralytics)
python3 -c "from ultralytics import YOLO; YOLO('yolov8n.pt').export(format='onnx', imgsz=640,
opset=17, simplify=True)"
cp yolov8n.onnx v1.0/public/models/yolov8n-objects.onnx

# 5. Dev server
npm run dev    # → http://localhost:9002

# 6. Chrome Extension (optional)
# chrome://extensions → Developer mode → Load unpacked → select extension/
```

Available Commands

```
npm run dev           # Dev server (Turbopack, port 9002)
npm run build         # Production build
npm run lint          # ESLint
npm run typecheck     # TypeScript type check
npm run genkit:dev    # Genkit AI dev server
```

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15. Configuration & Environment

```
NEXT_PUBLIC_FIREBASE_API_KEY=...
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN=...
NEXT_PUBLIC_FIREBASE_PROJECT_ID=...
NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET=...
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID=...
NEXT_PUBLIC_FIREBASE_APP_ID=...
GOOGLE_GENAI_API_KEY=...
```

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16. Performance Metrics

Metric	Value
Frame rate	5 FPS (~200ms intervals)
Face detection	TinyFaceDetector, 320px input, ~150ms
Memory	~50MB for face-api models
CPU usage	~15–25% on modern hardware
Latency target	< 200ms/frame
Object detection	~15ms/frame (WebGPU)

Accuracy Limitations

- Poor lighting reduces landmark accuracy
- Extreme head angles (>40°) lose tracking
- Reflective glasses affect gaze estimation
- Only first detected face is tracked (single-user)
- Low-res cameras reduce accuracy

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17. Privacy

1. **All processing local** — no video data sent to any server
2. **No raw video stored** — only derived scores persisted
3. **Extension data** — stored locally in Chrome's storage
4. **Firebase** — only attention scores + timestamps, never video

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18. References

- **EAR**: Soukupová & Čech (2016) — *Real-Time Eye Blink Detection using Facial Landmarks*
- **PERCLOS**: Dinges & Grace (1998) — *PERCLOS: A Valid Psychophysiological Measure of Alertness*
- **face-api.js**: https://github.com/vladmandic/face-api
- **68-Point Landmarks**: iBUG 300-W dataset standard
- **YOLOv8**: Ultralytics (2023) — https://github.com/ultralytics/ultralytics

- **MediaPipe Hands**: Google (2020) — https://developers.google.com/mediapipe

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19. Cross-Reference: v2.0

v1.0's per-person analysis pipeline is **reused as the core of v2.0**. See the v2.0 README for the multi-person CCTV server system.

Component	v1.0	v2.0
Face detection	face-api.js	YOLOv8-m + MediaPipe FaceMesh
68-pt landmarks	face-api.js	MediaPipe → ibug mapping
EAR/MAR/Gaze/HeadPose	TypeScript (browser)	Python port
Object detection	YOLOv8-nano (ONNX.js)	YOLOv8-s (Ultralytics)
Hand tracking	MediaPipe Hands (browser)	Pose covers this
Temporal classifier	1D-CNN (browser ONNX)	1D-CNN (server ONNX)
Tracking	Single person	ByteTrack multi-person
Seat mapping	N/A	Shapely polygon IoU
Deployment	Browser / Chrome Extension	FastAPI + GPU server